

# Nicolas Espinosa Dice

## Education

- 2023– Present **Ph.D., Computer Science, Cornell University**  
*Advisor:* Wen Sun  
*Research Areas:* Reinforcement Learning, Imitation Learning, Generative Models
- 2022 **B.S., Mathematics and Computer Science, Harvey Mudd College**  
*Advisors:* George D. Montanez and Dagan Karp  
Graduated with High Distinction and Departmental Honors in Mathematics and Computer Science.

## Research Experience

- 2023– Present **Department of Computer Science, Cornell University**  
*Advisor:* Wen Sun  
*Research Focus:* How can agents learn from both *interaction* and *offline data*: what can be done *offline* to better learn/explore/fine-tune *online*?
- 2020–2022 **AMISTAD Lab, Harvey Mudd College**  
*Advisor:* George D. Montanez  
Developed a probabilistic model of abductive reasoning, and derived generalization error bounds of learning algorithms in an algorithmic search framework.

## Industry Experience

- 2022–2023 **Research Scientist, Dasion**  
*Advisor:* Weiqing Gu  
Developed an audio-based model capable of detecting Autism Spectrum Disorder in children, contributing to a successful NSF SBIR Phase I Project Pitch, and co-authored successful NSF SBIR Phase II Proposal.
- 2021–2022 **Project Lead, Clinic Program, Harvey Mudd College**  
Project lead of undergraduate team working with Rockerbox that built an anomaly detection model on time-series marketing data.
- Summer 2021 **Software Engineer Intern, Etsy**  
Developed a transformer-based deep learning model with DistilBERT architecture using Tensorflow to classify safe search queries.
- Summer 2019 **Software Engineer Intern, Viasat**  
Built a heads-up display on a Microsoft HoloLens, integrated with data from Link 16 radio network. Presented by Viasat at AUSA 2019.

## Publications

- [1] **Nicolas Espinosa-Dice**, Yiyi Zhang, Yiding Chen, Bradley Guo, Owen Oertell, Gokul Swamy, Kianté Brantley, and Wen Sun. “Scaling Offline RL via Efficient and Expressive Shortcut Models”. In: *arXiv preprint arXiv:2505.22866* (2025).
- [2] **Nicolas Espinosa-Dice**, Sanjiban Choudhury, Wen Sun, and Gokul Swamy. “Efficient Imitation Under Misspecification”. In: *Preprint* (2024).
- [3] **Nicolas Espinosa-Dice**, Gokul Swamy, Sanjiban Choudhury, and Wen Sun. “Efficient Inverse Reinforcement Learning without Compounding Errors”. In: *First Reinforcement Learning Safety Workshop*. 2024.

- [4] Ramya Ramalingam, **Nicolas Espinosa-Dice**, Megan L Kaye, and George D Montañez. “Bounding Generalization Error Through Bias and Capacity”. In: *2022 International Joint Conference on Neural Networks (IJCNN)*. IEEE. 2022, pp. 1–8.
- [5] **Nicolas Espinosa-Dice**, Megan Kaye, Hana Ahmed, and George Montañez. “A Probabilistic Theory of Abductive Reasoning”. In: *Proceedings of the 13th International Conference on Agents and Artificial Intelligence*. Vol. 2. 2021.

## Teaching

- 2024 **Teaching Assistant**, *Foundations of Reinforcement Learning*, Cornell University
- 2021 **Grader and Tutor**, *Artificial Intelligence*, Harvey Mudd College

## Outreach and Reviewing

- 2024 **Conference Reviewer**, International Conference on Learning Representations
- 2024 **ASCEND Mentor**, *Department of Computer Science*, Cornell University
- 2024 **Graduate Student Mentor**, *Department of Computer Science*, Cornell University
- 2021 **Judiciary Chair**, *Honor Board*, Harvey Mudd College
- 2020 **Public Outreach Director**, *Society of Professional Latinx in STEM*, Harvey Mudd College

## Fellowships

- 2025 **NSF Graduate Research Fellowship Program**, *National Science Foundation*
- 2025 **Bowers-CIS LinkedIn Fellowship** (Declined), *LinkedIn*
- 2023 **Hopper-Dean and Bowers Fellowship**, *Cornell University*
- 2018 **Harvey S. Mudd Merit Award**, *Harvey Mudd College*

## Awards

- 2023 **Dean’s Scholar**, *Cornell University*
- 2022 **Departmental Honors in Computer Science**, *Harvey Mudd College*
- 2022 **Departmental Honors in Mathematics**, *Harvey Mudd College*
- 2022 **Don Chamberlain Computer Science Research Award**, *Harvey Mudd College*  
Second student at Harvey Mudd to win both Chamberlain Research and Clinic Program Individual Awards
- 2022 **Clinic Program Individual Award**, *Harvey Mudd College*